

Flow Separation Criterion

Continuity Equation (curvilinear)

$$\frac{1}{1+Kn} \frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} + \frac{K}{1+Kn} (\ell u_n) = 0$$

At wall, no slip and no permeability.

$$(u_s, u_n) \Big|_{n=0, \text{wall}} = (0, 0)$$

→ At $n=0$, wall, s = any value in the domain

$$u_s(s, n=0) \Rightarrow u_s(s) \Big|_{n=0} = 0$$

$$u_n(s, n=0) \Rightarrow u_n(s) \Big|_{n=0} = 0$$

Due to no slip

$$\text{Therefore, } \frac{\partial u_s}{\partial s}(s) \Big|_{n=0} = 0 \text{ and } \frac{\partial u_n}{\partial s}(s) \Big|_{n=0} = 0$$

$$\rightarrow \left(\frac{1}{1+K(0)} \frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} + \frac{K}{1+K(0)} (\ell u_n) \right) \Big|_{n=0} = 0$$

$$\left(\frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} \right) \Big|_{n=0} = 0$$

$$\left(\ell \frac{\partial u_s}{\partial s} + u_s \frac{\partial \ell}{\partial s} + \ell \frac{\partial u_n}{\partial n} + u_n \frac{\partial \ell}{\partial n} \right) \Big|_{n=0} = 0$$

$$\therefore \left(\ell \frac{\partial u_n}{\partial n} \right) \Big|_{n=0} = 0; \ell > 0 \text{ at all } s$$

$$\text{Therefore, } \frac{\partial u_n}{\partial n} \Big|_{n=0} = 0 \text{ at wall for all } s$$

Flow Separation Criterion

Streamwise/tangential momentum equation (curvilinear)

$$\rho \left(\frac{u_s}{1+Kn} \frac{\partial u_s}{\partial s} + \frac{\partial u_s}{\partial n} + \frac{K u_s u_n}{1+Kn} \right) = - \frac{1}{1+Kn} \frac{\partial p}{\partial s} + \frac{1}{1+Kn} \frac{\partial \tau_{ss}}{\partial s}$$

$$0 = - \frac{\partial p}{\partial s} + \frac{\partial \tau_{ss}}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K \tau_{sn} \text{ at wall} + \frac{\partial \tau_{sn}}{\partial n} + \frac{2K}{1+Kn} \tau_{sn}$$

$$\tau_{ss} = \frac{4}{3} \mu \left(\frac{1}{1+Kn} \frac{\partial u_s}{\partial s} + \frac{K u_n}{1+Kn} \right) - \frac{2}{3} \mu \frac{\partial u_n}{\partial n} \text{ at wall } (n=0, u_s=0, u_n=0)$$

$$\frac{\partial \tau_{ss}}{\partial s} = \frac{4}{3} \mu \left(\frac{1}{1+Kn} \frac{\partial^2 u_s}{\partial s^2} + \frac{\partial u_s}{\partial s} \frac{\partial}{\partial s} \left(\frac{1}{1+Kn} \right) + \frac{\partial u_n}{\partial s} \left(\frac{K}{1+Kn} \right) + \frac{\partial}{\partial s} \left(\frac{K}{1+Kn} \right) \right)$$

$$u_s(s) \Big|_{n=0} = 0 \rightarrow \frac{\partial u_s(s)}{\partial s} \Big|_{n=0} = 0 \rightarrow \frac{\partial^2 u_s(s)}{\partial s^2} \Big|_{n=0} = 0 - \frac{2}{3} \mu \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0}$$

$$u_n(s) \Big|_{n=0} = 0 \rightarrow \frac{\partial u_n(s)}{\partial s} \Big|_{n=0} = 0 \left[\frac{\partial \tau_{ss}}{\partial s} \right] = - \frac{2}{3} \mu \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0}$$

$$\frac{\partial u_n(s)}{\partial n} \Big|_{n=0} = 0 \rightarrow \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0} = 0$$

$$\therefore \frac{\partial \tau_{ss}}{\partial s} = - \frac{2}{3} \mu(0) \rightarrow \frac{\partial \tau_{ss}}{\partial s} = 0$$

$$\therefore \left(0 = - \frac{\partial p}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K \tau_{sn} \right) \Big|_{n=0, \text{ wall}}$$

Flow Separation Criterion

$$\tau_{sn} = \mu \left(\frac{\partial u_s}{\partial n} - \frac{K u_s}{1+Kn} + \frac{1}{1+Kn} \frac{\partial u_n}{\partial s} \right)$$

$u_s=0$ $\frac{\partial u_n}{\partial s}=0$

$$\tau_{sn} = \mu \frac{\partial u_s}{\partial n} \Big|_{n=0} \text{ at wall because}$$

$$\therefore \left(0 = -\frac{\partial p}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K\mu \frac{\partial u_s}{\partial n} \right) \Big|_{n=0} \quad \begin{matrix} u_s(s)=0 \text{ and } u_n(s)=0 \\ \text{and } \frac{\partial u_n(s)}{\partial s} = 0 \end{matrix}$$

$$\frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - \frac{K}{1+Kn} \frac{\partial u_s}{\partial n} - \frac{1}{1+Kn} \frac{\partial}{\partial n} \left(\frac{1}{1+Kn} \frac{\partial u_n}{\partial s} \right) \right) \Big|_{n=0}$$

$$\therefore \frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - \frac{K}{1+K(0)} \frac{\partial u_s}{\partial n} + \frac{1}{1+K(0)} \frac{\partial^2 u_n}{\partial n \partial s} \right) \Big|_{n=0}$$

$$\frac{\partial^2 u_n}{\partial n \partial s} = \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right); \quad \frac{\partial^2 u_n}{\partial n \partial s} (s, n=0) = \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \Big|_{n=0} \right) = 0$$

$$\frac{\partial u_n(s)}{\partial n} \Big|_{n=0} = 0 \rightarrow \frac{\partial}{\partial s} \left(\frac{\partial u_n(s)}{\partial n} \Big|_{n=0} \right) = 0$$

$$\rightarrow \left(\frac{\partial \tau_{sn}}{\partial n} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - K \frac{\partial u_s}{\partial n} \right) \right) \Big|_{n=0}$$

$$\frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0} - K\mu \frac{\partial u_s}{\partial n} \Big|_{n=0}$$

Flow Separation Criterion

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} + 2K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} - K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0} + 2K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} + K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\boxed{\left.\frac{\partial p}{\partial s}\right|_{n=0} = \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} + K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}}$$

①

Flow Separation Criterion

$$\text{Since } \left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + K\mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

Let's nondimensionalize n and u_s (variables)

$$n' = \frac{n}{\delta} \quad \text{and} \quad u'_s = \frac{u_s}{u_{s,e}}$$

$$\text{When } n' = 0, \frac{n}{\delta} = 0 \rightarrow n = 0(\delta) \rightarrow n = 0$$

When $n' = 0, n = 0$. Therefore, $u_s(n=0) = 0$ when $n' = 0$

Since $u'_s = 0$ because $n' = 0$, $u'_s = \frac{0}{u_{s,e}} = 0$ (because $u_{s,e}$ is nonzero) when $n' = 0$

So because of $n' = 0, u'_s = 0$

$$\text{When } n' = 1, \frac{n}{\delta} = 1 \rightarrow n = 1 \cdot \delta \rightarrow n = \delta$$

When $n' = 1, n = \delta$. Therefore, $u_s(n=\delta) = u_{s,e}$ when $n' = 1$

Since $u_s = u_{s,e}$ because $n' = 1$, $u'_s = \frac{u_{s,e}}{u_{s,e}} = 1$ when $n' = 1$

So because of $n' = 1, u'_s = 1$

Therefore, $(n', u'_s) \in \{(n' = 0, u'_s = 0), (n' = 1, u'_s = 1)\}$

Flow Separation Criterion

Now, $(n', u'_s) \in \{(n'=0, u'_s=0), (n'=1, u'_s=1)\}$

Domain of n : $0 \leq n < \delta$

Since $n' = \frac{n}{\delta}$, $0 \leq \frac{n}{\delta} < 1 \Rightarrow 0 \leq n' < 1$ is the domain of n'

Therefore, Taylor series with respect to n' will

converge which is the reason for why I

non-dimensionalized n and u_s to n' and u'_s

respectively to use Taylor series approximation for the entire boundary layer which converges ($n' < 1$, so $n' \downarrow$ to 0)

Taylor Series Approximation (3rd order)

$$u'_s(n') = u'_s \Big|_{n'=0} + n' \frac{\partial u'_s}{\partial n'} \Big|_{n'=0} + \frac{(n')^2}{2!} \frac{\partial^2 u'_s}{\partial n'^2} \Big|_{n'=0} + \frac{(n')^3}{3!} \frac{\partial^3 u'_s}{\partial n'^3} \Big|_{n'=0} + O\left(\frac{n'^4}{4!}\right)$$

where $0 \leq n' < 1$, therefore $\frac{(n')^4}{24} \approx 0$ as an approximation

$$O\left(\frac{n'^4}{4!}\right) \approx 0,$$

$$\therefore u'_s = n' \frac{\partial u'_s}{\partial n'} \Big|_{n'=0} + \frac{(n')^2}{2} \frac{\partial^2 u'_s}{\partial n'^2} \Big|_{n'=0} + \frac{(n')^3}{6} \frac{\partial^3 u'_s}{\partial n'^3} \Big|_{n'=0}$$

Flow Separation Criterion

Now,

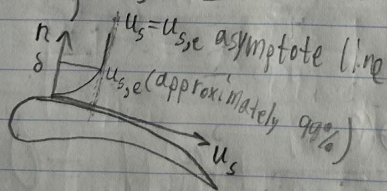
$$u_s' = n' \frac{\partial u_s'}{\partial n'} \Big|_{n'=0} + \frac{(n')^2}{2} \frac{\partial^2 u_s'}{\partial n'^2} \Big|_{n'=0} + \frac{(n')^3}{6} \frac{\partial^3 u_s'}{\partial n'^3} \Big|_{n'=0}$$

At $n'=1$, $u_s'=1$ and

$$1 = (1) \frac{\partial u_s'}{\partial n'} \Big|_{n'=0} + \frac{1^2}{2} \frac{\partial^2 u_s'}{\partial n'^2} \Big|_{n'=0} + \frac{1^3}{6} \frac{\partial^3 u_s'}{\partial n'^3} \Big|_{n'=0}$$

$$\rightarrow \frac{\partial u_s'}{\partial n'} = \frac{\partial u_s'}{\partial n'} \Big|_{n'=0} + n' \frac{\partial^2 u_s'}{\partial n'^2} \Big|_{n'=0} + \frac{(n')^2}{2} \frac{\partial^3 u_s'}{\partial n'^3} \Big|_{n'=0}$$

At $n'=1$, $u_s' \rightarrow 1$ like an asymptote.



Therefore, $\frac{\partial u_s'}{\partial n'} \Big|_{n=\delta} = 0$

Since $u_s' = \frac{u_s}{u_{s,e}} \Rightarrow u_s = u_{s,e} u_s'$, $n' = \frac{n}{\delta} \Rightarrow \frac{\partial n'}{\partial n} = \frac{1}{\delta}$

$$\frac{\partial u_s}{\partial n} = \frac{\partial (u_{s,e} u_s')}{\partial n} = u_{s,e} \frac{\partial u_s'}{\partial n'} \frac{\partial n'}{\partial n} = \frac{u_{s,e}}{\delta} \frac{\partial u_s'}{\partial n'}$$

Since $\frac{\partial u_s}{\partial n} = \frac{u_{s,e}}{\delta} \frac{\partial u_s'}{\partial n'}$

When $\frac{\partial u_s}{\partial n} \Big|_{n=\delta} = 0 \rightarrow \frac{u_{s,e}}{\delta} \frac{\partial u_s'}{\partial n'} \Big|_{n'=1} = 0$, $u_{s,e}$ and δ are nonzero
Therefore, $\frac{\partial u_s'}{\partial n'} \Big|_{n'=1} = 0$

Flow Separation Criterion

Since

$$\frac{\partial u'_s}{\partial n'} = \frac{\partial u'_s}{\partial n'} \bigg|_{n'=0} + n' \frac{\partial^2 u'_s}{\partial n'^2} \bigg|_{n'=0} + \frac{(n')^2}{2} \frac{\partial^3 u'_s}{\partial n'^3} \bigg|_{n'=0}$$

Since

$$\frac{\partial u'_s}{\partial n'} \bigg|_{n'=1} = 0$$

Therefore, substituting $n'=1$

$$\frac{\partial u'_s}{\partial n'} \bigg|_{n'=1} = 0 = \frac{\partial u'_s}{\partial n'} \bigg|_{n'=0} + (1) \frac{\partial^2 u'_s}{\partial n'^2} \bigg|_{n'=0} + \frac{(1)^2}{2} \frac{\partial^3 u'_s}{\partial n'^3} \bigg|_{n'=0}$$

$$0 = \frac{\partial u'_s}{\partial n'} \bigg|_{n'=0} + \frac{\partial^2 u'_s}{\partial n'^2} \bigg|_{n'=0} + \frac{1}{2} \frac{\partial^3 u'_s}{\partial n'^3} \bigg|_{n'=0}$$

$$\frac{\partial^3 u'_s}{\partial n'^3} \bigg|_{n'=0} = -2 \frac{\partial u'_s}{\partial n'} \bigg|_{n'=0} - 2 \frac{\partial^2 u'_s}{\partial n'^2} \bigg|_{n'=0}$$

Flow Separation Criterion

Since, $\left. \frac{\partial^3 u_s'}{\partial n'^3} \right|_{n'=0} = -2 \left. \frac{\partial u_s'}{\partial n'} \right|_{n'=0} - 2 \left. \frac{\partial^2 u_s'}{\partial n'^2} \right|_{n'=0}$

And since when $n'=1$, from $u_s' = n' \left. \frac{\partial u_s'}{\partial n'} \right|_{n'=0} + \frac{(n')^2}{2} \left. \frac{\partial^2 u_s'}{\partial n'^2} \right|_{n'=0} + \frac{(n')^3}{6} \left. \frac{\partial^3 u_s'}{\partial n'^3} \right|_{n'=0}$

$$u_s'(n'=1) = 1 = (1) \left. \frac{\partial u_s'}{\partial n'} \right|_{n'=0} + \frac{1^2}{2} \left. \frac{\partial^2 u_s'}{\partial n'^2} \right|_{n'=0} + \frac{1^3}{6} \left. \frac{\partial^3 u_s'}{\partial n'^3} \right|_{n'=0}$$

Substitute,

$$1 = \left. \frac{\partial u_s'}{\partial n'} \right|_{n'=0} + \frac{1}{2} \left. \frac{\partial^2 u_s'}{\partial n'^2} \right|_{n'=0} + \frac{1}{6} \left(-2 \left. \frac{\partial u_s'}{\partial n'} \right|_{n'=0} - 2 \left. \frac{\partial^2 u_s'}{\partial n'^2} \right|_{n'=0} \right)$$

$$1 = \frac{2}{3} \left. \frac{\partial u_s'}{\partial n'} \right|_{n'=0} + \frac{1}{6} \left. \frac{\partial^2 u_s'}{\partial n'^2} \right|_{n'=0} \quad (2)$$

Since $\left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + K \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} \quad (1)$

Let's nondimensionalize the equation 1

$$\left. \frac{\partial u_s}{\partial n} \right|_{n=0} = \frac{u_{s,e}}{\delta} \left. \frac{\partial u_s'}{\partial n'} \right|_{n'=0} \text{ was already derived}$$

$$\frac{\partial^2 u_s}{\partial n^2} = \frac{\partial}{\partial n} \left(\frac{\partial u_s}{\partial n} \right) = \frac{\partial}{\partial n} \left(\frac{u_{s,e}}{\delta} \frac{\partial u_s'}{\partial n'} \right) = \frac{\partial}{\partial n'} \left(\frac{u_{s,e}}{\delta} \frac{\partial u_s'}{\partial n'} \right) \frac{\partial n'}{\partial n}$$

Flow Separation Criterion

Since $\frac{\partial^2 u_s}{\partial n^2} = \frac{\partial}{\partial n'} \left(\frac{u_{s,e}}{\delta} \frac{\partial u'_s}{\partial n'} \right) \frac{\partial n'}{\partial n}$, and since $n' = \frac{n}{\delta}$
so $\frac{\partial n'}{\partial n} = \frac{1}{\delta}$

$$\frac{\partial^2 u_s}{\partial n^2} = \frac{\partial}{\partial n'} \left(\frac{u_{s,e}}{\delta} \frac{\partial u'_s}{\partial n'} \right) \left(\frac{1}{\delta} \right) = \frac{u_{s,e}}{\delta^2} \frac{\partial^2 u'_s}{\partial n'^2}$$

$$\text{So } \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} = \frac{u_{s,e}}{\delta^2} \left. \frac{\partial^2 u'_s}{\partial n'^2} \right|_{n'=0}$$

$$\text{and } \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = \frac{u_{s,e}}{\delta} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0}$$

$$\text{Since } \left. \frac{\partial P}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + K\mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

Substitute,

$$\left. \frac{\partial P}{\partial s} \right|_{n=0} = \frac{\mu u_{s,e}}{\delta^2} \left. \frac{\partial^2 u'_s}{\partial n'^2} \right|_{n'=0} + \frac{K\mu u_{s,e}}{\delta} \left. \frac{\partial u'_{s,e}}{\partial n'} \right|_{n'=0}$$

Flow Separation Criterion

Since $\left. \frac{\partial P}{\partial s} \right|_{n=0} = \frac{\mu u_{s,e}}{\delta^2} \left. \frac{\partial^2 u'_s}{\partial n'^2} \right|_{n'=0} + \frac{\kappa \mu u_{s,e}}{\delta} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0}$

And $1 = \frac{2}{3} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} + \frac{1}{6} \left. \frac{\partial^2 u'_s}{\partial n'^2} \right|_{n'=0}$

$6 = 4 \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} + \left. \frac{\partial^2 u'_s}{\partial n'^2} \right|_{n'=0}$

$\left. \frac{\partial^2 u'_s}{\partial n'^2} \right|_{n'=0} = 6 - 4 \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0}$

Substitute

$\rightarrow \left. \frac{\partial P}{\partial s} \right|_{n=0} = \frac{\mu u_{s,e}}{\delta^2} \left(6 - 4 \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} \right) + \frac{\kappa \mu u_{s,e}}{\delta} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0}$

$\left. \frac{\partial P}{\partial s} \right|_{n=0} = \frac{6\mu u_{s,e}}{\delta^2} - \frac{4\mu u_{s,e}}{\delta^2} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} + \frac{\kappa \mu u_{s,e}}{\delta} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0}$

$\left. \frac{\partial P}{\partial s} \right|_{n=0} = \frac{6\mu u_{s,e}}{\delta^2} - \frac{4\mu u_{s,e}}{\delta^2} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} + \frac{\kappa \mu u_{s,e}}{\delta^2} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0}$

Flow Separation Criterion

Since $\left. \frac{\partial P}{\partial s} \right|_{n=0} = \frac{6\mu u_{s,e}}{\delta^2} - \frac{4\mu u_{s,e}}{\delta^2} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} + \frac{K\delta\mu u_{s,e}}{\delta^2} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0}$

$$\therefore \left. \frac{\partial P}{\partial s} \right|_{n=0} + \frac{4\mu u_{s,e}}{\delta^2} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} - \frac{K\delta\mu u_{s,e}}{\delta^2} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} = \frac{6\mu u_{s,e}}{\delta^2} \quad (3)$$

$$\left. \frac{\partial P}{\partial s} \right|_{n=0} + \left(\frac{4}{\delta} - \frac{K\delta}{\delta} \right) \left(\mu \right) \frac{u_{s,e}}{\delta} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} = \frac{6\mu u_{s,e}}{\delta^2}$$

$$\frac{u_{s,e}}{\delta} \left. \frac{\partial u'_s}{\partial n'} \right|_{n'=0} = \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\left. \frac{\partial P}{\partial s} \right|_{n=0} + \left(\frac{4}{\delta} - \frac{K\delta}{\delta} \right) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = \frac{6\mu u_{s,e}}{\delta^2}$$

Multiply both sides by δ^2

$$\delta^2 \left. \frac{\partial P}{\partial s} \right|_{n=0} + (4\delta - K\delta^2) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = 6\mu u_{s,e}$$

$$(4\delta - K\delta^2) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = 6\mu u_{s,e} - \delta^2 \left. \frac{\partial P}{\partial s} \right|_{n=0}$$

Flow Separation Criterion

$$(4\delta - K\delta^2) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = 6\mu u_{s,e} - \delta^2 \left. \frac{\partial p}{\partial s} \right|_{n=0}$$

$$\therefore \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = \frac{6\mu u_{s,e} - \delta^2 \left. \frac{\partial p}{\partial s} \right|_{n=0}}{4\delta - K\delta^2}$$

Since $\mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = \tau_{\text{wall}}$

$$\tau_{\text{wall}} = \frac{6\mu u_{s,e} - \delta^2 \left. \frac{\partial p}{\partial s} \right|_{n=0}}{4\delta - K\delta^2}$$

For flow to separate, the airflow must no longer exert shear force on the wall and shear force reverses and becomes negative.

$\tau_{\text{wall}} \leq 0$ for flow to separate

Flow Separation Criterion

$\tau_{wall} \leq 0$ for flow to separate

$$\frac{6\mu u_{s,e} - \delta^2 \frac{\partial p}{\partial s} \big|_{n=0}}{4\delta - K\delta^2} = \tau_{wall} \leq 0$$

$$\therefore \frac{6\mu u_{s,e} - \delta^2 \frac{\partial p}{\partial s} \big|_{n=0}}{4\delta - K\delta^2} \leq 0$$

for flow to separate

For flow to stay attached or reattach,
the flow must be exerting positive
shear force on the wall. $\tau_{wall} > 0$.

$$\frac{6\mu u_{s,e} - \delta^2 \frac{\partial p}{\partial s} \big|_{n=0}}{4\delta - K\delta^2} = \tau_{wall} > 0$$

$$\therefore \frac{6\mu u_{s,e} - \delta^2 \frac{\partial p}{\partial s} \big|_{n=0}}{4\delta - K\delta^2} > 0$$

for flow to stay attached or reattach